# 1. Differentiation

## 1.1. Basic Derivatives

| | |
|---|---|
| $x^n$ | $nx^{n-1}$ |
| $e^u$ | $\frac{\mathrm{du}}{\mathrm{dx}}e^u$ |
| $\ln u$ | $\frac{\frac{\mathrm{du}}{\mathrm{dx}}}{u}$ |
| $\sin(ax)$ | $a\cos(ax)$ |
| $\cos(ax)$ | $-a\sin(ax)$ |
| $\tan(ax)$ | $a\sec^2(ax)$ |
| $tan^{-1}(ax)$ | $\frac{a}{1+(ax)^2}$ |

## 1.2. Chain, Product and Quotient Rule

- Chain Rule:

$$\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{dy}}{\mathrm{du}} \times \frac{\mathrm{du}}{\mathrm{dx}}$$

- Product Rule:

$$\frac{d}{\mathrm{dx}}(\mathrm{uv}) = u\frac{\mathrm{dv}}{\mathrm{dx}} + v\frac{\mathrm{du}}{\mathrm{dx}}$$

- Quotient Rule:

$$\frac{d}{\mathrm{dx}}\left(\frac{u}{v}\right) = \frac{v\frac{\mathrm{du}}{\mathrm{dx}} - u\frac{\mathrm{dv}}{\mathrm{dx}}}{v^2}$$

## 1.3. Parametric Equations

$$\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{dy}}{\mathrm{dt}} \div \frac{\mathrm{dx}}{\mathrm{dt}}$$

- In a parametric equation $x$ and $y$ are given in terms of $t$ and you must use the above rule to find the derivative

## 1.4. Implicit Functions

- These represent circles or lines with circular curves, on a Cartesian plane
- Difficult to rearrange in form $y = \therefore$ differentiate as is
- Differentiate $x$ terms as usual
- For $y$ terms, differentiate the same as you would $x$ but multiply with $\frac{\mathrm{dy}}{\mathrm{dx}}$
- Then make $\frac{\mathrm{dy}}{\mathrm{dx}}$ the subject of formula for derivative



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